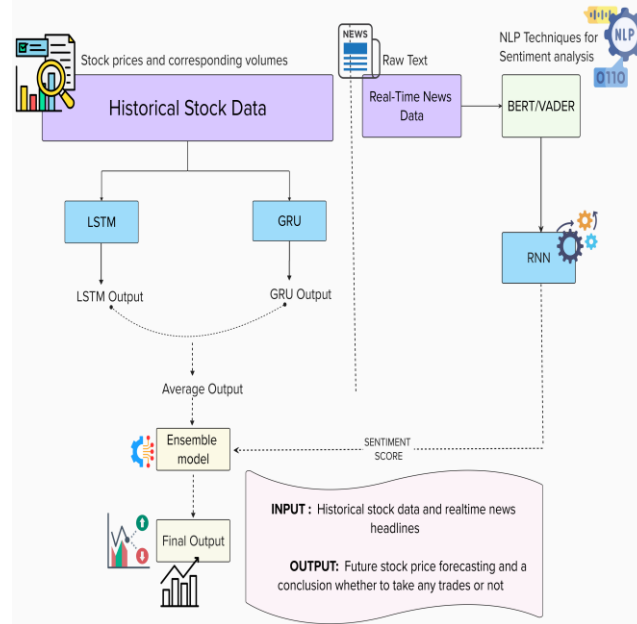


Sentiment-Infused Stock Price Prediction: A Hybrid Model for Capturing Market Sentiments and Financial Trends

Problem Statement

The stock market's rapid fluctuations and complex influences make real-time prediction difficult. Traditional models face challenges like scalability and delays, requiring more advanced solutions for accurate forecasting.

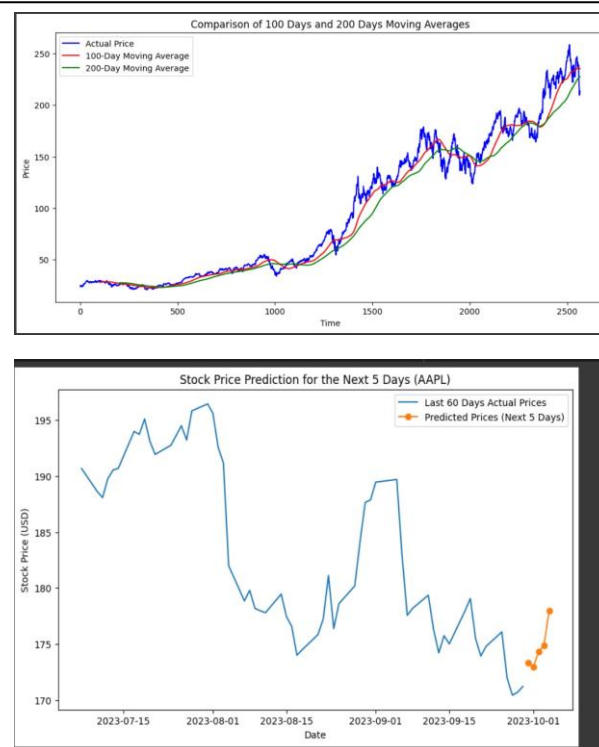
System Architecture



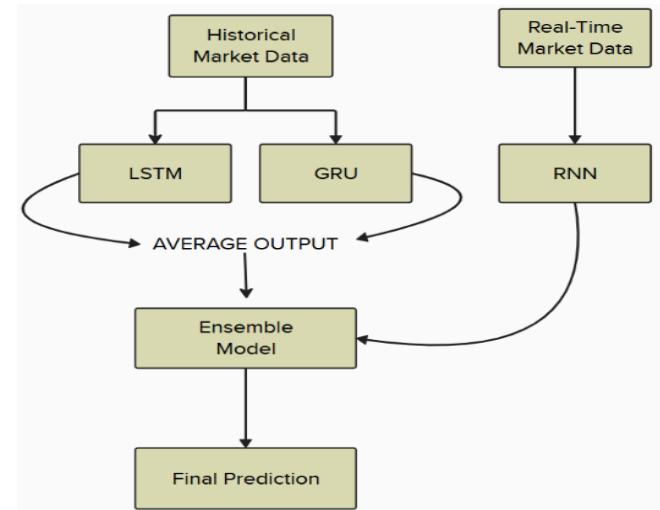
Objectives

1. Develop hybrid LSTM-GRU models for stock forecasting.
2. Build an ensemble model combining LSTM, GRU, and RNN.
3. Use BERT and VADER for sentiment-based stock prediction.
4. Compare deep learning and traditional models for accuracy and risk.

Result



Work Flow



Key References

- [1] J. Liu, Z. Wang, and B. Zheng, "A Deep Reinforcement Learning-Based Decision Support System for Automated Stock Market Trading," IEEE Access, vol. 8, pp. 143676-143686, 2020.
- [2] Sheikh Mohammad Idrees, M. Afshar Alam, and Parul Agarwal, "A Prediction Approach for Stock Market Volatility Based on Time Series Data," IEEE Access, vol. 7, pp. 17287-17295, 2019.

Conclusion

The system combines LSTM, GRU, and sentiment analysis to enhance stock prediction accuracy and trading strategies, even in volatile markets.